

Machine Vision – Week 1 Lab Report

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Task A – OpenCV Basics

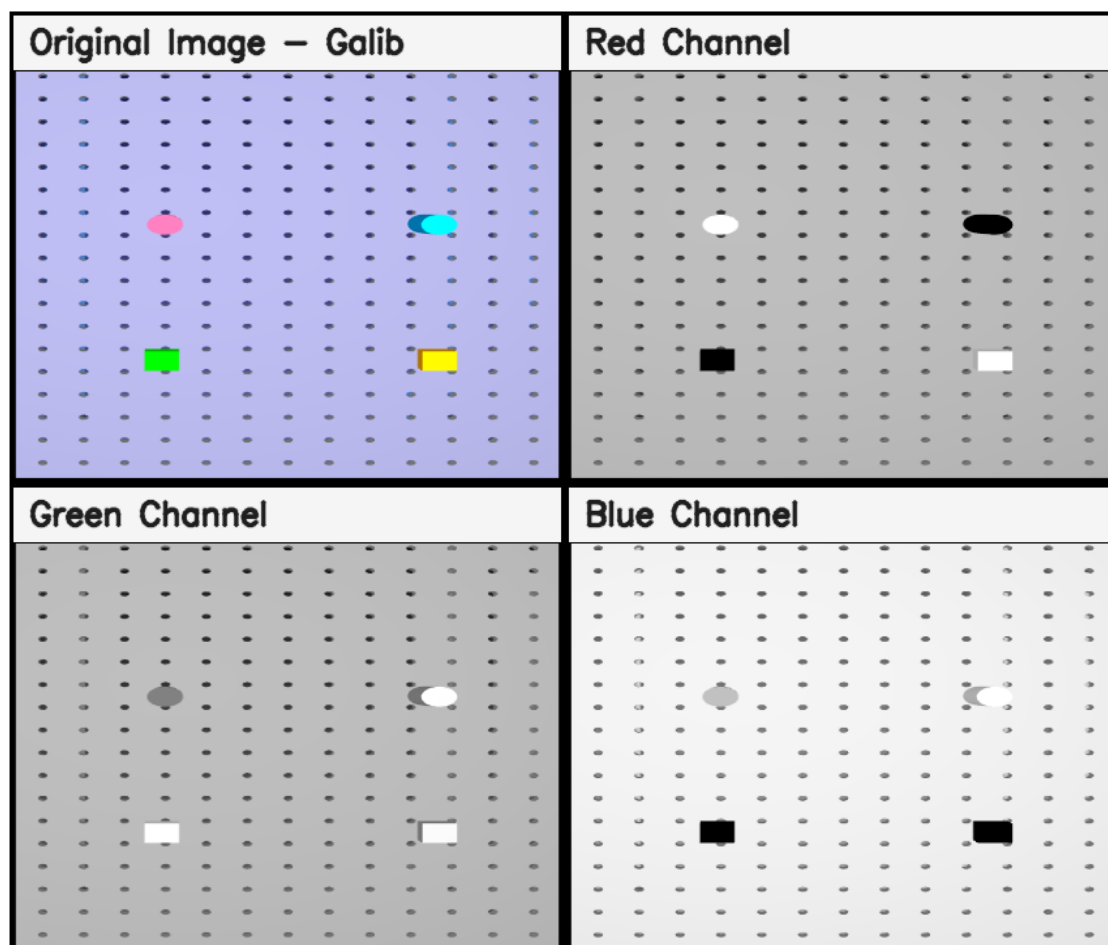
In Task A, I used OpenCV to process an image captured from the RoboDK simulated environment. The image was loaded using OpenCV, which reads images in BGR format. I then separated the image into its individual color channels (Blue, Green, and Red) using the `cv2.split()` function. Each channel was converted into a displayable format so that they could be visualized clearly.

To organize the results, I created a 2×2 grid that includes:

- The original image (with my name added)
- The red channel
- The green channel
- The blue channel

This task helped me understand how different color channels contribute to the overall image and how OpenCV represents color information.

Task A Output



Task B – RoboDK + OpenCV Annotation

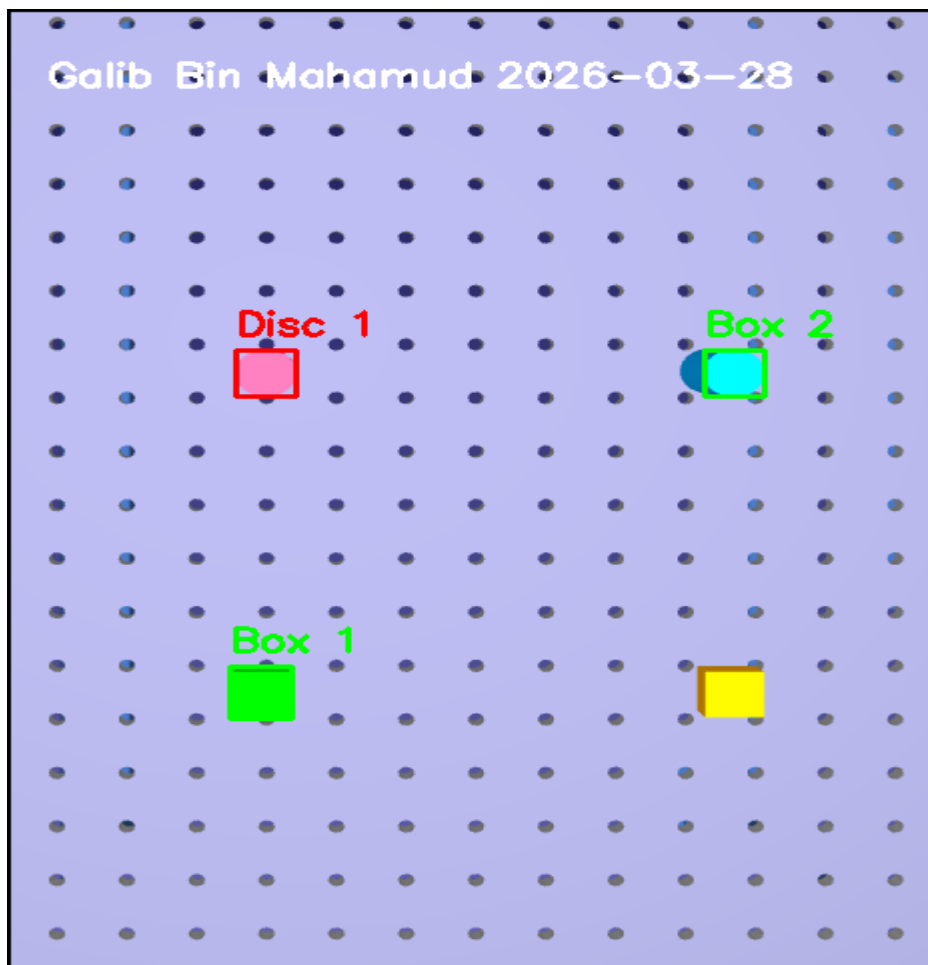
In Task B, I used OpenCV to detect objects from an image captured from the RoboDK simulation. The image was first converted from BGR to HSV color space to make color-based detection more effective. I created color masks to identify specific objects in the scene.

In particular:

- Red color was used to detect the disc
- Green color was used to detect the boxes

To improve detection accuracy, I applied morphological operations to remove small noise from the masks. Contours were then detected from the processed masks, and bounding rectangles were drawn around the identified objects. Each detected object was labeled accordingly. Finally, I added my name and the current date at the top of the image and saved the result as annotated.png.

Task B Output



Findings and Learning Experience

- Learned difference between BGR and RGB
- Understood RGB channel separation
- Used HSV color space for object detection
- Applied contour detection to identify objects
- Learned image annotation using OpenCV